

Yixuan Hua

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Education

Princeton University, Princeton, NJ, United States

Aug 2022 - Present

Ph.D. in Operations Research & Financial Engineering

Advisor: Amir Ali Ahmadi, Bartolomeo Stellato

Peking University, Beijing, China

Sep 2018 - July 2022

B.S. in Mathematics, Overall GPA: 3.82/4.0, Major Rank: 5/37

Advisor (undergraduate research): Zaiwen Wen

Professional Experience

J.P. Morgan

Jersey City, NJ

Quantitative Analytics Program – Summer Associate

June 2026 - Aug 2026

- Reviewed and validated a loss-given-default (LGD) model for CRE credit risk, performing independent analytical checks and proposing benchmark models to evaluate performance and risk drivers.
- Built an L1-regularization variable selection workflow, producing candidate models with stronger performance and comparable robustness.
- Developed a survival analysis model for workout-period timing risk; used predicted survival probability as a new LGD risk driver, improving model performance.
- Built a Python web UI prototype for repeatable, structured comparison across candidate models.

Flexcar

Boston, MA

Strategic Intern – Finance

June 2025 - Aug 2025

- Built a logistic regression model to analyze customer conversion and identified key behavioral drivers of signup-to-first-order completion.
- Built a survival analysis model to analyze customer tenure and price sensitivity, quantifying how pricing changes affected retention.
- Developed an individualized pricing strategy using models' estimates of conversion and tenure to optimize revenue and customer lifetime value.
- Analyzed order velocity using SARIMA and evaluated its relationship with inventory size and distribution.

Publications

Journal Articles

Overparameterized ReLU Neural Networks Learn the Simplest Model: Neural Isometry and Phase Transitions.

Y. Wang, **Y. Hua**, E. J. Candès and M. Pilanci.

IEEE Transactions on Information Theory, doi: 10.1109/TIT.2025.3530355.

Preprints

Disjunctive Sum of Squares.

A. A. Ahmadi, S. Dash, **Y. Hua**, B. Stellato.

arXiv e-prints: 2605.28674, 2026.

Talks

"Disjunctive Sum of Squares"

INFORMS, Seattle, 2024

ICCOPT, University of Southern California, 2025

"The most often profitable portfolio"

MOPTA, Lehigh University, 2026

Teaching Experiences

Convex and Conic Optimization

Spring 2024, 2025, 2026

Professors: A. A. Ahmadi (Spring 2024, 2025), I. Akrotirianakis (Spring 2026)

Computing and Optimization for the Physical and Social Sciences

Fall 2023, 2024, 2025

Professor: A. A. Ahmadi

Honors and Awards

Merit Student, 2018-2019 academic year

The Suzhou Industrial Park Scholarship, 2018-2019 academic year

First Prize, The Jiang Zehan Mathematical Contest in Modeling, 2019

First Prize, The Mathematics Competition of Chinese College Students, 2019

First Prize in Zhejiang Province, National High School Mathematics League, 2016 and 2017

Silver Medal, China Mathematics Olympiad, 2016

Skills

Programming: Python, R, MATLAB, Julia

Optimization Software: CVXPY, Gurobi, Mosek

Other Tools: LaTeX, Git

Languages: English (fluent), Mandarin Chinese (native)